Solid State for Devices Physics
Tutorial Problems and Solutions

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1 Basic Review Questions

Semiconductor Materials

Question: List two elemental semiconductor materials and two compound semiconductor materials. **Answer:**

- **Elemental semiconductors:** Silicon (Si), Germanium (Ge)
- **Compound semiconductors:** Gallium Arsenide (GaAs), Indium Phosphide (InP)

[Other examples: Silicon Carbide (SiC), Gallium Nitride (GaN)]

Lattice Structures

Question: Sketch three lattice structures: (a) simple cubic, (b) body-centered cubic, and (c) face-centered cubic. **Answer:**

- **Simple Cubic (SC):** Atoms at 8 corners of cube only. (1 atom/unit cell).
- **Body-Centered Cubic (BCC):** Atoms at 8 corners + 1 at center. (2 atoms/unit cell).
- **Face-Centered Cubic (FCC):** Atoms at 8 corners + 6 face centers. (4 atoms/unit cell).

[Note: Most semiconductors have diamond or zinc-blende structures (based on FCC)]

Volume Density of Atoms

Question: Describe the procedure for finding the volume density of atoms in a crystal. **Answer: Procedure**

1. Identify the crystal structure and lattice constant a.
2. Count the number of atoms per unit cell n:
   - Corner atoms contribute 1/8 each
   - Face atoms contribute 1/2 each
   - Body-centered atoms contribute 1
3. Calculate unit cell volume: $V = a^3$ (for cubic).
4. Compute volume density:

$$N_V = \frac{n}{V} = \frac{n}{a^3} \quad (\text{atoms/cm}^3)$$

Miller Indices

Question: Describe the procedure for obtaining the Miller indices that describe a plane in a crystal. **Answer: Procedure for Miller Indices** (hkl)

1. Find where the plane intercepts the x, y, z axes.
2. Express intercepts as multiples of lattice constants (pa, qa, ra).
3. Normalize to get (p, q, r).
4. Take reciprocals: $(1/p, 1/q, 1/r)$.
5. Clear fractions by multiplying by the Least Common Multiple (LCM).
6. Result is (h, k, l) — the Miller indices.

[Note: If a plane is parallel to an axis, that index is 0. Negative indices are written with a bar: $(\bar{1}23)$.]

Impurities in Crystals

Question: What is meant by a substitutional impurity in a crystal? What is meant by an interstitial impurity? **Answer: Substitutional Impurity:**

- A foreign atom replaces a host atom at a lattice site.
- Example: Phosphorus (P) or Boron (B) replacing Si in silicon crystal (used for doping).

Interstitial Impurity:

- A foreign atom occupies the space between lattice sites.
- Example: Small atoms like H, C, or O in silicon (generally causes lattice strain).

Wave–Particle Duality

Question: State the wave–particle duality principle and state the relationship between momentum and wavelength. **Answer: Wave–Particle Duality Principle:** All matter and radiation exhibit both wave-like and particle-like properties. **de Broglie Relationship:**

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

where λ is wavelength, $p = mv$ is momentum, and $h \approx 6.626 \times 10^{-34}\ \mathrm{J \cdot s}$ is Planck's constant.

Quantized Energy Levels

Question: What is meant by quantized energy levels? Can an electron in a potential well have an arbitrary energy? **Answer: Quantized Energy Levels:** In confined systems (atoms, quantum wells), electrons can only occupy specific discrete energy values, not a continuous range. **Answer to second part: NO.** An electron in a potential well cannot have arbitrary energy. It can only have specific allowed energies E_n.

$$\text{Example (infinite well): } E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}, \quad n = 1, 2, 3, \ldots$$

2 Schrödinger Equation Review

Schrödinger Equation: Basic Form

Question: Write down the time-dependent Schrödinger equation. What physical quantity does the wavefunction $\Psi(x,t)$ describe? **Answer: Time-Dependent Schrödinger Equation (1D):**

$$i\hbar\frac{\partial\Psi(x,t)}{\partial t} = \hat{H}\Psi(x,t) = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x,t)\right]\Psi(x,t)$$

Physical Meaning of $\Psi(x,t)$: The wavefunction itself has no direct physical meaning, but $|\Psi(x,t)|^2$ represents the **probability density** of finding the particle at position x at time t.

Separation of Variables

Question: Explain why we can write $\Psi(x,t) = \psi(x)\phi(t)$ when the potential is time-independent. What form does $\phi(t)$ take? **Answer: Why separation works:** When the potential V is time-independent, the time and spatial dependencies in the Schrödinger equation separate completely, allowing us to write solutions as products. **Form of $\phi(t)$:**

$$\phi(t) = e^{-iEt/\hbar}$$

The full solution is $\Psi(x,t) = \psi(x)e^{-iEt/\hbar}$, where E is the energy eigenvalue and $\psi(x)$ satisfies the time-independent Schrödinger equation.

Stationary States

Question: What does it mean for a wavefunction to be a *stationary state*? Why is the probability density $|\Psi(x,t)|^2$ time-independent in this case? **Answer: Stationary State:** A state with definite energy E where all measurable properties are time-independent. The wavefunction has the form $\Psi(x,t) = \psi(x)e^{-iEt/\hbar}$. **Why probability density is time-independent:**

$$|\Psi(x,t)|^2 = \Psi^*\Psi = \left(\psi^*(x)e^{+iEt/\hbar}\right)\cdot\left(\psi(x)e^{-iEt/\hbar}\right) = |\psi(x)|^2$$

The time-dependent phase factors cancel out, leaving only the time-independent spatial part.

Infinite Square Well

Question: For a particle in a 1D infinite square well of width a:

What is the general form of the wavefunction inside the well? State the boundary conditions at $x = 0$ and $x = a$. Write the expression for the allowed energy levels E_n.

Answer:

- **Wavefunction inside the well ($0 < x < a$):**

$$\psi_n(x) = \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi x}{a}\right), \quad n = 1, 2, 3, \ldots$$

- **Boundary conditions:**

$$\psi(0) = 0 \quad \text{and} \quad \psi(a) = 0$$

- **Allowed energy levels:**

$$E_n = \frac{n^2\pi^2\hbar^2}{2ma^2} = \frac{n^2h^2}{8ma^2}, \quad n = 1, 2, 3, \ldots$$

Free Particle

Question: Write down the solution of the Schrödinger equation for a free particle of energy $E = \frac{\hbar^2 k^2}{2m}$. What is the physical meaning of the parameter k? **Answer: Solution for free particle:**

$$\Psi(x,t) = Ae^{i(kx-\omega t)} \quad \text{or} \quad \Psi(x,t) = Ae^{ikx}e^{-iEt/\hbar}$$

Physical meaning of k: k is the **wave number** (wave vector in 1D), related to wavelength λ and momentum p:

- $k = 2\pi/\lambda$
- $p = \hbar k$

Finite Barrier

Question: If a particle has energy $E < V_0$ in a finite potential barrier, what form does the wavefunction take inside the barrier? **Answer: Wavefunction inside the barrier ($E < V_0$):**

$$\psi(x) = Ce^{\kappa x} + De^{-\kappa x}$$

where the decay constant is

$$\kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

This represents **exponential decay**, leading to the phenomenon of quantum tunneling.

Normalisation

Question: Why must the total wavefunction satisfy

$$\int_{-\infty}^{+\infty} |\Psi(x,t)|^2 dx = 1 \ ?$$

Answer: $|\Psi(x,t)|^2$ represents the **probability density** of finding the particle at position x at time t. Since the particle must exist *somewhere* in space with certainty (probability = 1), the integral of the probability density over all space must equal 1. This is the **normalization condition**.

3 Arrhenius Relationship

Arrhenius Relation for Thermal Defects (General Formula)

The general formula for defect concentration n_d as a function of temperature T is:

$$n_d = N \exp\left(-\frac{E_f}{k_B T}\right)$$

where $k_B \approx 8.617 \times 10^{-5}\,\text{eVK}^{-1}$ is the Boltzmann constant.

Vacancy Fraction at High Temperature

Given: $N = 8.5 \times 10^{22}\,\text{cm}^{-3}$, $E_f = 1.2\,\text{eV}$, $T = 1500\,\text{K}$. **Solution:**

$$f = \exp\left(-\frac{E_f}{k_B T}\right) = \exp\left(-\frac{1.2}{8.617 \times 10^{-5}\,\text{eVK}^{-1} \times 1500\,\text{K}}\right) \approx e^{-9.28} \approx 9.3 \times 10^{-5}$$

$$n_d = Nf = 8.5 \times 10^{22}\,\text{cm}^{-3} \times 9.3 \times 10^{-5} \approx 7.9 \times 10^{18}\,\text{cm}^{-3}$$

Answer: The fractional concentration is $f \approx 9.3 \times 10^{-5}$, and the density is $n_d \approx 7.9 \times 10^{18}\,\text{cm}^{-3}$.

Estimating E_f from Two Concentrations

Given: $n_1 = 2.0 \times 10^{15}\,\mathrm{cm}^{-3}$ at $T_1 = 1000\,\mathrm{K}$; $n_2 = 4.0 \times 10^{16}\,\mathrm{cm}^{-3}$ at $T_2 = 1200\,\mathrm{K}$. **Solution:** Using the ratio $\frac{n_2}{n_1} = \exp\left[-\frac{E_f}{k_B}(\frac{1}{T_2} - \frac{1}{T_1})\right]$, we solve for E_f:

$$E_f = k_B \frac{\ln(n_2/n_1)}{1/T_1 - 1/T_2}$$

$$\ln\frac{n_2}{n_1} = \ln(20) \approx 3.00$$

$$\frac{1}{T_1} - \frac{1}{T_2} \approx 1.667 \times 10^{-4}\,\mathrm{K}^{-1}$$

$$E_f \approx \frac{(8.617 \times 10^{-5}\,\mathrm{eVK}^{-1})(3.00)}{1.667 \times 10^{-4}\,\mathrm{K}^{-1}} \approx 1.55\,\mathrm{eV}$$

Answer: $E_f \approx 1.6\,\mathrm{eV}$

Effect of Entropy Factor

Given: $E_f = 2.5\,\mathrm{eV}$, $S_f/k_B = 1.5$, $N = 6.0 \times 10^{22}\,\mathrm{cm}^{-3}$, $T = 1000\,\mathrm{K}$. **Solution:** The defect concentration n_d changes by a factor of $A = \exp(S_f/k_B)$.

- Entropy factor $A = \exp(1.5) \approx 4.48$
- Exponent: $E_f/(k_BT) \approx 29.0$
- Defect concentration without entropy ($A = 1$):

$$n_d(A = 1) = 6.0 \times 10^{22} \times e^{-29.0} \approx 1.54 \times 10^{10}\,\mathrm{cm}^{-3}$$

- Defect concentration with entropy ($A = 4.48$):

$$n_d = 1.54 \times 10^{10} \times 4.48 \approx 6.9 \times 10^{10}\,\mathrm{cm}^{-3}$$

Answer: n_d rises from $\approx 1.5 \times 10^{10}\,\mathrm{cm}^{-3}$ to $\approx 6.9 \times 10^{10}\,\mathrm{cm}^{-3}$.

Temperature for Given Vacancy Density

Given: $N = 1.0 \times 10^{23}\,\mathrm{cm}^{-3}$, $E_f = 1.8\,\mathrm{eV}$, $n_d = 1.0 \times 10^{15}\,\mathrm{cm}^{-3}$. **Solution:**

- Fraction $f = n_d/N = 1 \times 10^{-8}$
- From $f = \exp(-E_f/k_BT)$, we solve for T: $T = \frac{E_f}{k_B \ln(1/f)}$
- $\ln(1/f) = \ln(1 \times 10^8) \approx 18.42$
- $T = \frac{1.8\,\mathrm{eV}}{8.617 \times 10^{-5}\,\mathrm{eVK}^{-1} \times 18.42} \approx 1130\,\mathrm{K}$

Answer: $T \approx 1130\,\mathrm{K}$

Comparing Two Materials

Given: Material A: $E_f = 2.0\,\mathrm{eV}$; Material B: $E_f = 2.8\,\mathrm{eV}$; $T = 1200\,\mathrm{K}$, $N = 7.0 \times 10^{22}\,\mathrm{cm}^{-3}$.
Solution: $k_BT \approx 0.1034\,\mathrm{eV}$

- **Material A:** $E_f/(k_BT) \approx 19.3$. $n_d^A \approx 2.9 \times 10^{14}\,\mathrm{cm}^{-3}$
- **Material B:** $E_f/(k_BT) \approx 27.0$. $n_d^B \approx 1.3 \times 10^{11}\,\mathrm{cm}^{-3}$

Answer: $n_A \approx 2.9 \times 10^{14}\,\mathrm{cm}^{-3}$, $n_B \approx 1.3 \times 10^{11}\,\mathrm{cm}^{-3}$. The ratio $n_A/n_B \approx 2200$.

4 Crystal Structure

Miller Indices and Interplanar Spacing

Given: Simple cubic crystal with lattice constant $a = 0.30\,\text{nm}$. A plane intersects the axes at $(x, y, z) = (2a, a, a/2)$. **Part (i): Find Miller Indices** (hkl)

1. Normalized intercepts in units of a: $(2, 1, 1/2)$
2. Reciprocals: $(\frac{1}{2}, 1, 2)$
3. Clear fractions (multiply by 2): $(1, 2, 4)$

Miller indices: $(hkl) = (124)$. **Part (ii): Find Interplanar Spacing** d_{hkl} For cubic crystals:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} = \frac{0.30\,\text{nm}}{\sqrt{1^2 + 2^2 + 4^2}} = \frac{0.30\,\text{nm}}{\sqrt{21}} \approx 0.0655\,\text{nm}$$

Answer: $(hkl) = (124)$, $d_{124} \approx 0.0655\,\text{nm}$.

5 Quantum Mechanics Exercises

Photon Energy and Wavelength

Problem: Calculate the energy of each photon of blue light of frequency $\nu = 6.40 \times 10^{14}\,\text{Hz}$. What is the wavelength of this photon? **Solution:**

- **Part 1: Energy of photon** ($E = h\nu$, using $h \approx 4.136 \times 10^{-15}\,\text{eV} \cdot \text{s}$)

$$E = (4.136 \times 10^{-15}\,\text{eV} \cdot \text{s}) \times (6.40 \times 10^{14}\,\text{s}^{-1}) \approx 2.64\,\text{eV}$$

- **Part 2: Wavelength** ($\lambda = c/\nu$, using $c \approx 2.998 \times 10^8\,\text{m/s}$)

$$\lambda = \frac{2.998 \times 10^8\,\text{m/s}}{6.40 \times 10^{14}\,\text{s}^{-1}} \approx 468\,\text{nm}$$

Answer: $E \approx 2.64\,\text{eV}$, $\lambda \approx 468\,\text{nm}$.

de Broglie Wavelength

Problem:

- Calculate the de Broglie wavelength of an electron moving at $v = 1 \times 10^5\,\text{m/s}$.
- Calculate the de Broglie wavelength of a person (mass = 70 kg) walking at 1 m/s.

Solution ($\lambda = h/mv$):

- **Electron:**

$$\lambda_e = \frac{6.625 \times 10^{-34}\,\text{J} \cdot \text{s}}{(9.11 \times 10^{-31}\,\text{kg}) \times (1 \times 10^5\,\text{m/s})} \approx 7.27 \times 10^{-9}\,\text{m} = 7.27\,\text{nm}$$

- **Person:**

$$\lambda_p = \frac{6.625 \times 10^{-34}\,\text{J} \cdot \text{s}}{(70\,\text{kg}) \times (1\,\text{m/s})} \approx 9.46 \times 10^{-36}\,\text{m}$$

Answer and Interpretation: $\lambda_{\text{electron}} \approx 7.27\,\text{nm}$. $\lambda_{\text{person}} \approx 9.46 \times 10^{-36}\,\text{m}$. The wavelength of the macroscopic object is negligible.

Heisenberg Uncertainty Principle

Problem: The uncertainty in the position of an electron is $\Delta x = 8$ Å.

- Determine the minimum uncertainty in the momentum Δp.
- Calculate the uncertainty in the kinetic energy (assume $p \approx \Delta p$).

Solution:

- **Part 1: Minimum Uncertainty in Momentum** ($\Delta p_{\text{min}} = \hbar/(2\Delta x)$)

$$\Delta p_{\text{min}} = \frac{1.0546 \times 10^{-34}\,\text{J} \cdot \text{s}}{2 \times 8 \times 10^{-10}\,\text{m}} \approx 6.59 \times 10^{-26}\,\text{kg} \cdot \text{m/s}$$

- **Part 2: Uncertainty in Kinetic Energy** ($\Delta E_K \approx \Delta p^2/(2m)$)

$$\Delta E_K \approx \frac{(6.59 \times 10^{-26}\,\text{kg} \cdot \text{m/s})^2}{2 \times 9.11 \times 10^{-31}\,\text{kg}} \approx 2.38 \times 10^{-21}\,\text{J} \approx 0.015\,\text{eV}$$

Answer: $\Delta p_{\text{min}} \approx 6.59 \times 10^{-26}\,\text{kg} \cdot \text{m/s}$ and $\Delta E_K \approx 0.015\,\text{eV}$.